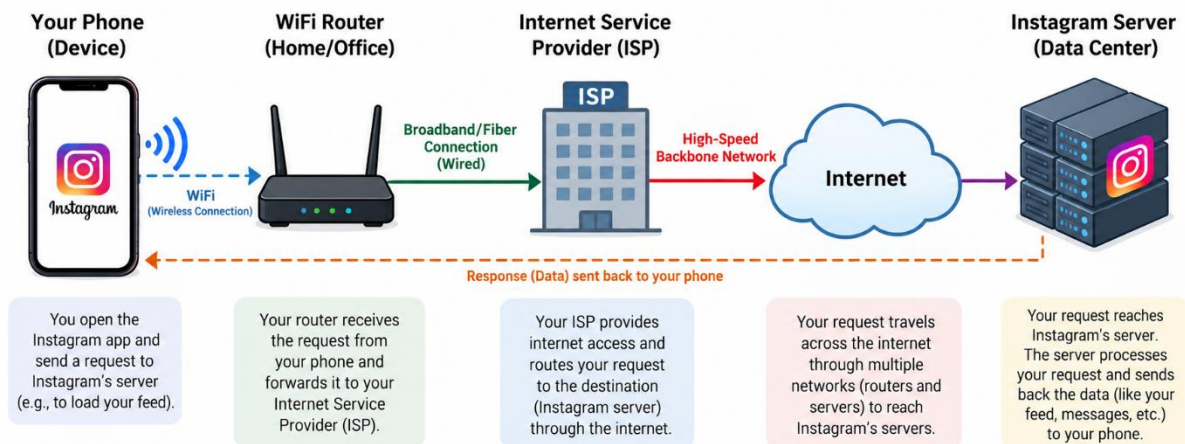


Introduction to Networks

1. Draw a simple diagram (hand-drawn or digital) showing how your phone connects to Instagram, including your device, WiFi router, ISP, and Instagram server. Label each part and the connections.



2. List 5 devices you use or see daily that are part of a network (for example: smartphone, smart TV, WiFi router, laptop, smartwatch). For each, write what type of network it connects to (LAN, WAN, PAN, etc.).

No.	Device	Network Type	Example
1	Smartphone	WAN	Connects to the internet using mobile data (4G/5G).
2	Laptop	LAN	Connects to a home/office network through Wi-Fi.
3	Smart TV	LAN	Connects to the home Wi-Fi network to access Netflix, YouTube, etc.
4	Wi-Fi Router	LAN / WAN	Connects devices within the home LAN and connects the LAN to the internet (WAN).
5	Smartwatch	PAN	Connects to a smartphone using Bluetooth, which is a Personal Area Network.

3. Explain the difference between the Internet and an intranet using an example from your college or a public WiFi you have used.

The **Internet** is a public network that connects computers and devices around the world. It allows users to access websites and online services such as Google, YouTube, Instagram, and Gmail. An **Intranet** is a private network used by an organization, such as a college or company, to share information and resources only with authorized users. For example, when I connect my laptop to my **college Wi-Fi**, I can use the Internet to browse websites and access online services. If the college has a private portal, attendance system, notices, or other resources that can only be accessed by students and staff through the college network, those resources are part of the **Intranet**. In simple words, the **Internet is a public network**, while an **Intranet is a private network used within an organization**.

4. Pick any one app you use (like WhatsApp, Zomato, or Flipkart) and describe, in 4-5 lines, how data travels from your device to their server and back when you perform a search inside the app.

Example: Instagram

When I search for something on **Instagram**, my smartphone sends the search request through **Wi-Fi to the Wi-Fi router**. The router forwards the request through my **ISP (Internet Service Provider)** and the Internet to Instagram's **server**. The Instagram server processes my search and finds the relevant accounts, posts, or content. The results are then sent back through the Internet, ISP, and Wi-Fi router to my smartphone, where they are displayed in the app.